

Pilot Log — Session 8, 3 August 2026

Entries 090–097 · Author: Ken Tombs (operator) with Claude (design lead, Opus 4.8) · The first BUILD session: the Phase 2 mediation layer went from paper (ADR-004) to running, connected infrastructure. Role inversion noted — the design lead produced the artefact (server code); the operator deployed and ran it. Save to GIT Deliverables.

1. Session summary

Session 8 built and connected the Dossier Space MCP Custody Server v1 — the laboratory bench of ADR-003/ADR-004 — taking Phase 2 from specification to executable, verified infrastructure. Python 3.14.6 and the MCP SDK were installed; the SDK's newest major version (2.0.0) proved to have dropped the API the server targets, and was deliberately pinned back to the documented 1.29.0 line — the pilot's stability-over-currency doctrine applied to its own toolchain. The six-tool server (all read-and-verify, running in Phase 2a passthrough) was written against the frozen ADR, loaded and ran standalone, was registered with Claude Code, and verified "V Connected". The subject window was retitled for the two-window discipline. The one outstanding item — the subject-side smoke test proving a task flows through the pipe — was blocked not by any architectural fault but by an expired subject login, hypothesised to follow a prior switch of the subject's authentication to Anthropic credits. Session banked at a clean point; the smoke test carries to the next session (Entry 098 reserved for its result).

2. Log entries

Entry 090 — Python runtime installed — a logged configuration act

Python 3.14.6 + pip 26.1.2 installed on the workstation via the official signed python.org installer (signature verified), under the user profile, callable via the py launcher. Recorded as a DELIBERATE, logged configuration change per the frozen-subject discipline that all configuration changes are logged acts — not a silent installer side-effect. PATH deliberately NOT modified (the launcher suffices), which later avoided the most common MCP stdio failure by design.

Entry 091 — MCP SDK installed; toolchain proven — and its versions pinned in the record

The MCP Python SDK installed cleanly on 3.14.6 (initially mcp 2.0.0 with its dependency tree). Core versions recorded for reproducibility — the kind of methods-section detail that is maddening to reconstruct later. Also confirmed: the bleeding-edge Python did NOT block the install, resolving the risk flagged at Session 8 opening.

Entry 092 — Custody Server placed outside the trust boundary

Server code located at E:\AAII XPlain\Dossier Space Pilot\Custody Server — at the pilot top level, deliberately OUTSIDE the subject's trust boundary (enron_mail_20150507). The subject reaches the server's TOOLS via MCP but can never see its SOURCE as files: the bench must not sit on the subject's floor. The server is configured to READ the evidence while being invisible to the subject — that asymmetry is the design.

Entry 093 — SDK API break found — pinned to the documented line (stability over currency, applied to the SDK)

The server failed to load: `ModuleNotFoundError` on `mcp.server.fastmcp`. Inspection showed SDK 2.0.0 had reorganised its API and DROPPED the FastMCP convenience interface the server (and all current documentation) targets. Caught at first load by the deliberate bare-run smoke test, before any integration effort was invested. Resolved by a logged downgrade: `pip install "mcp<2" → mcp 1.29.0`, the late stable 1.x release with FastMCP. Rationale, on the record: for evidence infrastructure that must be assured, reviewed, and reproducible, the WELL-DOCUMENTED version beats the NEWEST version — the exact reasoning by which the subject was frozen and "stable" was chosen over "latest". Currency is not a virtue in evidence infrastructure; reproducibility is. The build phase's first act demonstrated the pilot's own thesis on itself: the tool built to govern a fast-moving subject is itself subject to fast-moving dependencies.

Entry 094 — The server loads and runs — the mediation layer exists as executable infrastructure

Post-pin, the server loaded and ran: both banner lines printed (`CUSTODY_MODE=2a`; evidence root correct), then it hung awaiting MCP input — correct behaviour for a stdio server. `Ctrl+C` produced a clean `KeyboardInterrupt` shutdown (verified as the expected stop, not a fault). Six tools registered under `mcp 1.29.0 / Python 3.14.6`, Phase 2a passthrough. For the first time the Phase 2 mediation layer exists as running code rather than paper. The server is written as EXPERIMENTAL MATERIAL (ADR-004 s7) — readable as much as runnable, tool descriptions frozen, custody machinery present but switched OFF by the 2a flag (Session 9 flips it to 2b with no rewrite).

Entry 095 — Registered with Claude Code and verified ✓ Connected

Registered via: `claude mcp add --transport stdio dossier-custody -- <abs-python-path> <server-path>`, run from the workspace so the server is scoped to the matter project (visible to the subject here, not leaking into unrelated work). ABSOLUTE Python path used deliberately — `PATH` was never modified, and the documented most-common stdio failure (app cannot find python on a narrower GUI `PATH`) was thereby avoided by design. Registration written to `C:\Users\ktomb\.claude.json`, alongside the frozen-subject config — the MCP registration is now part of the workstation's logged configuration state. `claude mcp list` reported: `dossier-custody ... ✓ Connected`. The handshake works: Claude Code launched its own copy of the server, spoke MCP, and received the tool list. The mediation layer is LIVE infrastructure.

Entry 096 — Two-window discipline made physical: the subject window titled

The subject was briefly launched in the bench window (the recurring window-role confusion). Corrected, and a durable fix adopted: the subject's window is titled "SUBJECT - Claude Code runs here" set in PowerShell BEFORE launch (Claude Code cannot be retitled once running). Paired with the AUDIT BENCH title, the two roles are now visually distinct in the taskbar. Procedure amendment (Wake-Up / Setup v1.0): title BOTH windows before use — a human-factors fix for the trap that has recurred throughout the pilot.

Entry 097 — Smoke test blocked by an expired subject login — auth added to what the freeze cannot hold

On a correct-floor launch (banner: v2.1.209, Sonnet 5, workspace path — three-part read passed), the smoke-test task could not run: the subject presented an expired-login prompt. No credentials were to hand. Operator notes a prior switch of the subject's authentication to ANTHROPIC CREDITS (API/Console billing, consistent with the "API Usage Billing" banner) rather than the account login — hypothesised (LIKELY, not confirmed: no timestamp alignment available) as the cause of the logout, rather than simple token expiry. The logout therefore traces to an operator/configuration change,

not the hiatus or any subject action — consistent with the pilot's robust asymmetry that disturbances originate with the humans and the vendor, never the subject.

Significance: authentication joins the model default (Entry 087) on the list of things the LOCAL freeze cannot hold (Entry 088) — the client binary and filesystem froze; the model default and the auth session, both server-side, did not. Both now require wake-up verification. Setup-procedure lesson gained, with a mechanism: changing the subject's authentication or billing mode is a SESSION-INVALIDATING configuration act and must be logged and re-verified like a model or primer change; an auth-mode switch made casually mid-matter can silently log the subject out and halt work — an availability failure at a bad moment. The design decision to place custody in server-side logging rather than the client's approval gate is reinforced: when the client stops asking — or stops authenticating — the infrastructure must still record. (Related observation held for Session 9: the smoke-test instruction, once issued, triggered NO per-command approval prompt for the MCP tool — Claude Code appears to gate bash and MCP differently; to be confirmed when the authenticated smoke test runs.)

3. Session outcome and what resumes

Session 8 achieved its substantive goal: the mediation layer is built, connected, and verified ✓ Connected. The subject-side smoke test — one task through the pipe — is the only outstanding item, blocked solely by the auth state. It resumes on account refresh (or Session 9 opening): re-authenticate the subject via the Console/credits route; three-part banner check (confirm Sonnet 5 — the roster has drifted once already this session); then the two-part list_folder smoke test proving a task flows through the server. Entry 098 is reserved for the result. Design-lead model this session: Opus 4.8 (the SDK pin and server code were well within range; no Fable escalation needed — the escalation remains reserved for Session 10).

4. Standing configuration state (for the next wake-up to verify)

Workstation: Python 3.14.6 + pip 26.1.2 (user profile; PATH unmodified); MCP SDK pinned at 1.29.0. Custody server at Custody Server\dossier_custody_server.py, CUSTODY_MODE=2a. Registered in .claude.json as dossier-custody (local scope, workspace project), ✓ Connected. Subject: v2.1.209, Sonnet 5 (re-pinned last session; verify again next launch); frozen-subject config active but bounded — model default and auth session are server-side and drift beyond it. Workspace: v0.1 deployed (CLAUDE.md 8,264 B); pilot_outputs empty; corpus untouched. Windows: AUDIT BENCH and SUBJECT titled. Open: the smoke test; GIT Deliverables dedupe; break-even note; procedure v1.0 reissues; the scorecard rubric.